Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

Biology

Class – 7

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Table of Contents

S. No	Chapter Name	Page. No.
1	Nutrition in Organisms	1-4
2	Respiration in Organisms	5-9
3	Adolescence	10-11
4	Reproduction in Plants	12-16
5	Soil	17-19
6	Our Forest	20-23

1 NUTRITION

Solutions

Worksheet – 1

1. **Correct answer – B**

   Holozoic nutrition involves the ingestion of complex organic substances, followed by internal digestion and absorption.

2. **Correct answer – C**

   These "false feet" are temporary extensions of the cell membrane that Amoebas use to surround and trap food particles.

3. **Correct answer – B**

   Since Amoebas are unicellular, all chemical breakdown of food occurs inside a food vacuole within that single cell.

4. **Correct answer – D**

   The human alimentary canal is a continuous muscular tube extending from the mouth to the anus, averaging about 30 feet or 9 meters in length.

5. **Correct answer – B**

   Canines are sharp, pointed teeth designed specifically for piercing and tearing through tough food like meat

6. **Correct answer – B**

   Peristalsis is the rhythmic contraction and relaxation of muscles in the digestive tract that moves food forward.

7. **Correct answer – C**

   Mucus forms a physical barrier that prevents the stomach's own highly acidic gastric juices from digesting its walls

8. **Correct answer – B**

   The liver synthesizes bile to help digest fats, but it is held in the gall bladder until needed in the small intestine.

9. **Correct answer – C**

   Villi are tiny, finger-like folds that maximize the space available for digested nutrients to pass into the bloodstream.

10. **Correct answer – B**

    The large intestine primarily recovers water and salts from undigested food material, turning waste into solid feces.

Worksheet – 2

1. **Correct answer – B**

   Masticated food is mixed with saliva to form a bolus, which is then pushed through the pharynx into the oesophagus.

2. **Correct answer – C**

   Digestion begins in the mouth where amylase breaks down complex starch into simple sugars.

3. **Correct answer – C**

   HCl creates the acidic environment necessary for the enzyme pepsin to become active and digest proteins

4. **Correct answer – B**

   After being churned in the stomach for 3-4 hours, the food becomes a pulp-like substance known as chyme.

5. **Correct answer – C**

   The duodenum is the upper part of the small intestine that receives bile juice specifically for the emulsification of fats.

6. **Correct answer – B**

   The large intestine does not secrete enzymes; its primary role is absorbing water and salts from waste.

7. **Correct answer – C**

In the rumen, food is partially digested and then returned to the mouth as "cud" for further chewing.

8. **Correct answer – B**

Grit acts as "internal teeth" within the muscular gizzard to pulverize hard food into a paste.

9. **Correct answer – D**

This two-step chewing process allows ruminants to extract the most nutrients possible from tough plant fibers.

10. **Correct answer – B**

Assimilation is the specific stage where nutrients are actually incorporated into the body's cell components.

Worksheet 3

1. **Correct answer – C**

Oxygen is a product of photosynthesis, not a requirement.

2. **Correct answer – B**

They use chemical energy (e.g., hydrogen sulphide) instead.

3. **Correct answer – C**

At night, plants stop photosynthesis and focus on respiration.

4. **Correct answer – C**

These are the specific "sucking" roots of parasites.

5. **Correct answer – B**

They grow in nitrogen-deficient soil and use insects as a supplement.

6. **Correct answer – A**

Unlike total parasites (Cuscuta), they still have some ability to function.

7. **Correct answer – C**

The algae (phycobiont) provide the food, while the fungus provides the home.

8. **Correct answer – B**

This is the most common bacteria involved in natural nitrogen fixation.

9. **Correct answer – C**

They perform "extra-cellular" digestion on decaying matter.

10. **Correct answer – A**

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – D**

The back (end part) of the tongue contains taste buds that are most sensitive to bitter substances.

2. **Correct answer – C**

In each adult jaw, there are 4 cutting incisors, 2 tearing canines, 4 grinding premolars, and 6 crushing molars.

3. **Correct answer – A**

Stomata are tiny pores on the leaf surface that facilitate the exchange of oxygen and carbon dioxide.

4. **Correct answer – B**

Bile performs emulsification, which breaks large fat globules into tiny droplets so enzymes can digest them more easily.

5. **Correct answer – C**

Also known as the Pitcher Plant, Nepenthes traps and digests insects to obtain nitrogen.

6. **Correct answer – A**

Plants capture light energy from the sun and store it as chemical energy in the form of glucose.

7. **Correct answer – B**

Amylase in saliva chemically converts complex, tasteless starch in bread into simpler, sweet-tasting sugars.

8. **Correct answer – B**

Lenticels are porous tissues found on the bark of woody stems that allow for gas exchange.

9. Correct answer – C

Chyme is the thick, acidic fluid consisting of partially digested food and gastric juices produced in the stomach.

10. Correct answer – A

Oxygen is produced as a byproduct when water molecules are split during the light-dependent reactions of photosynthesis.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – A, B, D

Villi are finger-like projections with thin walls and rich blood supplies; they do *not* have pores for food to pass through.

12. Correct answer – A, D

Green plants are autotrophs that produce food and release oxygen, while most animals are heterotrophs.

13. Correct answer – A, B

Amoeba secretes various digestive enzymes, including amylase (for carbohydrates) and proteinase (for proteins), into its food vacuole.

14. Correct answer – B, D

Carbon dioxide is a raw material used in photosynthesis, not an end product. The main end products of photosynthesis are glucose and oxygen.

15. Correct answer – A, B, C

Photosynthesis requires carbon dioxide, water, and sunlight (captured by chlorophyll) to produce glucose.

ASSERTION AND REASONING TYPE

16. Correct answer – A

17. Correct answer – A

18. Correct answer – A

The "cud" chewing process is specifically evolved to handle the complexity of plant-based diets.

19. Correct answer – A

This highlights the "Mutualism" aspect. The fungus provides the infrastructure, but without the algae's "kitchen," the system collapses.

MATRIX MATCHING TYPE

20. Correct answer – B

a-S, b-P, c-T, d-R, e-Q

21. Correct answer – D

a-T, b-P, c-S, d-Q, e-R

COMPREHENSHION TYPE

Passage1

22. Correct answer – C

23. Correct answer – B

24. Correct answer – C

Passage2

25. Correct answer – C

26. Correct answer – A

27. Correct answer – D

Previous Year Questions

1. Correct answer – B

In typical digestive diagrams, (ii) usually points to the liver; however, based on the text provided, bile is *stored* in the gallbladder and it is the liver that secretes it (and option B is often the "incorrect" trap if the label points to the stomach or pancreas).

2. Correct answer – D

This matches the ruminant chambers: Rumen (bacteria), Reticulum (cud), Omasum (water), and Abomasum (true stomach with enzymes).

3. Correct answer – D

Bile juice helps in the breakdown of fats into fatty acids and glycerol, not amino acids (which come from proteins).

4. Correct answer – A

Based on the list, Beans, Gram, and Lentil are the three leguminous plants (Pigeon pea, peanut, and chickpeas are other examples from your text).

5. Correct answer – B

Paramecium uses hair-like **cilia** to sweep food into its mouth-like opening, unlike Amoeba which uses pseudopodia.

6. Correct answer – A

All are false: (I) Chlorophyll is the pigment, (II) Mushrooms are saprotrophs, (III) The entire process is called Nutrition (digestion is just the breakdown).

7. Correct answer – D

In one jaw, there are 4 Incisors (X), 2 Canines (Z), and 4 Premolars (part of Y), totaling 10 (excluding the 6 molars).

8. Correct answer – A

Villi are the outgrowths in the small intestine, and the stomach secretes mucus, HCl, and digestive juices (pepsin).

9. Correct answer – C

Photosynthesis requires Sunlight, Water, Minerals, Chlorophyll, and Carbon Dioxide (Oxygen is a product, not a requirement).

10. Correct answer – B

Incisors are for biting/cutting, Molars for crushing/grinding, and Canines for tearing.

2 RESPIRATION

Solutions

Worksheet – 1

1. **Correct answer C) Aerobic respiration**
   Aerobic respiration requires oxygen to break down glucose into energy, carbon dioxide, and water. Without oxygen, this process cannot occur.
2. **Correct answer B) Inhalation**
   Inhalation is the process of drawing air into the lungs, which contains oxygen necessary for respiration.
3. **Correct answer C) Filter dust and harmful particles**
   Hairs and mucus trap dust, pollen, and other harmful particles from entering the respiratory system, protecting the lungs.
4. **Correct answer C) Ethanol, carbon dioxide, and energy**
   In the absence of oxygen, yeast ferments sugar to produce ethanol, carbon dioxide, and a small amount of energy.
5. **Correct answer B) Pharynx**
   The pharynx connects both the digestive and respiratory systems, allowing air to go to the lungs and food to go to the stomach.
6. **Correct answer C) Alveoli**
   Alveoli are tiny sac-like structures in the lungs where oxygen enters the blood and carbon dioxide leaves the blood.
7. **Correct answer B) 12–18 breaths per minute**
   At rest, the human adult typically breathes 12–18 times per minute to supply enough oxygen for the body's needs.
8. **Correct answer C) Haemoglobin**
   Haemoglobin is a protein in red blood cells that binds oxygen in the lungs and releases it to body tissues.
9. **Correct answer C) To speed up respiration and produce more energy to meet the body's demands**
   During exercise, muscles need more energy, so oxygen intake increases to speed up cellular respiration and ATP production.
10. **Correct answer B) Lactic acid**
    When oxygen is insufficient in muscles during intense activity, glucose is partially broken down into lactic acid and energy, causing temporary muscle fatigue.

WORKSHEET -2

1. **Correct answer C) Earthworm**
   Earthworms do not have lungs or gills. Oxygen diffuses directly through their moist skin into the blood, and carbon dioxide leaves the body the same way.
2. **Correct answer C) Both their skin and their lungs**
   Frogs can respire through their lungs when on land and through their moist skin, especially when underwater or during hibernation.
3. **Correct answer B) Fermentation**
   In the absence of oxygen, yeast ferments sugar to produce ethanol, carbon dioxide, and a small amount of energy.
4. **Correct answer D) Gills**
   Fish extract oxygen from water using gills, which are thin, feathery structures rich in blood vessels.
5. **Correct answer B) Specialized respiratory organs**
   Terrestrial animals breathe using Lungs or other specialized respiratory organs.
6. **Correct answer C) Cutaneous respiration**
   Breathing through the skin is called Cutaneous respiration.

7. **Correct Answer D) Cell membrane**
In organisms like Amoeba and Paramecium, gases diffuse through the cell membrane.
8. **Correct Answer B) Spiracles**
Insects breathe through tiny openings called Spiracles.
9. **Correct Answer C) Tracheae**
Spiracles open into a network of air tubes called Tracheae.
10. **Correct Answer B) Human**
Human uses Lungs for respiration.

Worksheet-3

1. **Correct Answer (C) Glucose**
Plants break down glucose (food) through oxidation to release energy. Oxygen is what helps the oxidation happen, but it isn't the energy source itself. Carbon dioxide and water are products, not sources.
2. **Correct Answer (D) All living cells**
Respiration is a vital process that happens in every living cell of a plant to keep it alive.
It is not limited to specific parts like leaves or roots, nor is it restricted to cells with chlorophyll (which are for photosynthesis).
3. **Correct Answer(C) Gaseous exchange**

   The primary role of stomata is to allow gases like carbon dioxide and oxygen to move in and out of the leaf.

   Roots absorb water, and chloroplasts prepare food.
4. **Correct answer (B) Oxygen**

   During aerobic respiration, glucose reacts with oxygen to release energy.

   Nitrogen is not used in this process, and carbon dioxide is a result of the reaction, not a starter.
5. **Correct answer (C) Growth, repair, and other life activities**

   Energy from respiration powers all essential functions, including cell division (growth) and fixing damaged tissues.
6. **Correct Answer (B) Carbon dioxide and water**

   The chemical "leftovers" of respiration that the plant releases are carbon dioxide and water.

   Glucose and oxygen are the "fuel" used up during the process.
7. **Correct Answer (C) Both day and night**

   Unlike photosynthesis, which needs light, respiration happens 24/7 because cells always need energy to stay alive. Plants do not stop breathing just because the sun goes down or the seasons change.
8. **Correct Answer(B) Air**

   Plants take in carbon dioxide from the surrounding air through their stomata.

   Roots and soil provide water and minerals, but not the gas needed for photosynthesis.
9. **Correct Answer (C) Soil**

   Plants use their root systems to absorb water from the soil.

   While some plants can absorb moisture from the air, the vast majority rely on their roots in the ground.
10. **Correct Answer (B) Guard cells**

    Two specialized guard cells surround each stoma; they swell or shrink to open and close the pore.

Competitive corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (B) To provide more oxygen to cells for faster energy release via respiration.**

   Intense physical activity like sprinting increases the demand for energy; breathing faster ensures more oxygen reaches the muscles to break down glucose. Hemoglobin levels don't decrease during a sprint, and lungs are designed to expand.

2. **correct answer (A) The diffusion gradient at the alveoli will decrease, making it harder to remove CO2 from the blood.**

   Gas exchange relies on moving from high to low concentration. If the room has high CO2, the "gap" between the blood's CO2 and the air's CO2 shrinks, making it difficult for the body to exhale waste.

   Your body cannot physically close your nostrils or switch hemoglobin to carry nitrogen.

3. **Correct Answer (C) Trachea**

   The trachea is the scientific name for the windpipe; its cartilaginous rings prevent it from collapsing. The esophagus is for food, the bronchus is a smaller branch, and the pharynx is the throat area.

4. **Correct Answer (B) Respiration**

   Respiration is specifically the process of breaking down food (glucose) to release energy.

   Photosynthesis makes the food; transpiration is water loss; absorption is taking in nutrients.

5. **Correct Answer (B) Respiration, because cells need energy to stay alive regardless of light**

   All living things must respire constantly to survive. A plant in the dark will stop photosynthesizing but will continue to "breathe" (respire).

   Photosynthesis requires active light, it cannot be "stored" for 48 hours.

6. **Correct Answer B) Oxidized**

   In cellular respiration, glucose reacts with oxygen (is oxidized) to release energy.

   Glucose is produced in photosynthesis and stored as starch, but energy release requires oxidation.

7. **Correct Answer (C) To allow dissolved oxygen to diffuse into the blood**

   Gills are packed with blood vessels (capillaries) to pick up oxygen from the water and dump carbon dioxide. Blood vessels are for gas transport, not for trapping food or regulating temperature in this context.

8. **Correct Answer(B) Humans use a circulatory system (blood) to transport gases, while cockroaches use a tracheal network to deliver air directly to cells**

   Cockroaches breathe through small holes called spiracles that lead to a network of tubes (tracheae) reaching every cell; humans rely on blood to carry oxygen from lungs to cells.

   Cockroaches do not have lungs on their legs, and humans do not breathe through their skin.

9. **Correct Answer(B) Higher, lower**

   Diffusion is the natural movement of particles from an area where there are many (higher concentration) to an area where there are fewer (lower concentration).

10. **Correct Answer (C) Air**

    Terrestrial (land-living) animals have evolved lungs or tracheal systems specifically to extract oxygen from the air.

    Water is for aquatic animals, and soil/moisture are not the primary sources of oxygen for most land animals.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer A) It involves oxidation of glucose, C) It releases energy, and D) It occurs in all living cells**

    Respiration is the process of oxidizing glucose to release energy (ATP) for the plant's survival.

    It is a universal life process that occurs in all living cells.

12. **Correct Answer A) Earthworms, B) Leeches, and C) Frogs**
Cutaneous respiration is breathing through the skin. Earthworms and leeches use this as their primary method. Frogs use their moist skin to breathe, especially when underwater or hibernating.
Cockroaches use a tracheal system (spiracles and air tubes), not their skin.
13. **Correct Answer A) Aerobic respiration and D) Respiration in fish**
Aerobic respiration explicitly requires oxygen to fully break down glucose. Fish use oxygen that is dissolved in water, which passes over their gills into their blood. Photosynthesis produces oxygen as a byproduct rather than using it as a raw material. Anaerobic respiration occurs specifically in the absence of oxygen.
14. **Correct Answer A) The plant's ability to take in Carbon Dioxide for photosynthesis, B) The plant's ability to release Oxygen as a byproduct, and C) The plant's ability to release energy from glucose via aerobic respiration**
Stomata are the primary ports for gaseous exchange. Blocking them stops from entering for photosynthesis and from entering for aerobic respiration. It also prevents waste (from photosynthesis) from leaving.
While stomata affect transpiration, water absorption (Option D) is primarily a function of the roots.
15. **A) Photosynthesis is an energy-storing process, while respiration is an energy-releasing process, B) During the day, the rate of photosynthesis is generally higher than the rate of respiration, and D) Both processes involve the exchange of gases through the stomata**

Photosynthesis stores solar energy in glucose bonds; respiration breaks those bonds to release energy. On a sunny day, plants typically produce more food via photosynthesis than they consume via respiration. Both rely on the stomata to move and in and out of the leaves. Statement C is incorrect; respiration happens all the time

ASSERTION AND REASONING TYPE

16. **Correct Answer A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**

When the diaphragm and ribs move (Reason), they increase the volume of the chest. This physical expansion causes the air pressure inside to drop, which pulls air in from the outside (Assertion).

17. **correct answer A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**

Because water holds very little dissolved oxygen compared to air (Reason), fish must pump water over their gills much faster (Assertion) to get the same amount of oxygen a human gets in one breath.

18. **Correct answer B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.**

It is true that respiration happens 24/7 (Assertion) and that it happens in all cells (Reason). however, the reason it happens at night is because life processes (like staying alive) never stop, not simply because the plant has many cells.

19. **Correct answer D) Assertion is false, but Reason is true.**

The Assertion is false because unicellular organisms are too simple to have organs like lungs or gills. The Reason is true because they rely entirely on simple diffusion through their cell membrane to breathe.

MATRIX MATCH TYPE

20. **Corect Answer (A) a – P, b – Q, c – R, d – S**
21. **Correct Answer (A) a – P, b – S, c – Q, d – R**

COMPREHENSION TYPE

Passage-1

22. **Correct Answer (B) Their skin must remain moist for gas exchange**
23. **Correct Answer (C) Anaerobic respiration**
24. **Correct Answer (B) Both photosynthesis and respiration will stop**

Passage-2

25. **Correct Answer (B) Nose → Trachea → Bronchi**
26. **Correct Answer (B) Oxygen**
27. **Correct Answer (C) Helps inhalation and exhalation**

Previous Year Questions

1. **Correct Answer (C) Food is broken down into carbon dioxide, water and energy in this process**

   This is the incorrect statement because it describes aerobic respiration.
   In anaerobic respiration, food is typically broken down into ethyl alcohol (ethanol) and carbon dioxide (in yeast) or lactic acid (in animal muscles), along with energy. It does release less energy and occurs without oxygen.
2. **Correct Answer (B) Emphysema-Infection in trachea**

   This is the incorrect match. Emphysema is a chronic condition where the alveoli (air sacs) in the lungs are damaged, usually due to smoking, not an infection of the trachea. Bronchitis is inflammation of the bronchi, and Flu is a viral infection of the respiratory tract.
3. **Correct Answer (A) Amoeba- Tracheal respiration**

   This is the incorrect match. Amoeba is a unicellular organism that performs respiration through simple diffusion across its cell membrane.
   Tracheal respiration is characteristic of insects like cockroaches. Salmon use gills (branchial), earthworms use skin (cutaneous), and penguins use lungs (pulmonary).
4. **Correct Answer (A) Nose → Trachea → Bronchi → Bronchioles → Alveoli**

   This is the correct sequence of air travel during inhalation. Air enters the nose, passes through the trachea, divides into the bronchi, branches into smaller bronchioles, and finally reaches the alveoli for gas exchange.
5. **Correct Answer (B)) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + Energy$**

   Aerobic respiration happens in the presence of oxygen

   Glucose ($C_6H_{12}O_6$) reacts with oxygen to release carbon dioxide, water, and energy that the body uses.

3 ADOLESCENCE

Solutions

Worksheet – 1

1. **Correct answer – B**
   This is the biologically defined range where a child transitions into adulthood.
2. **Correct answer – C**
   Limbs typically experience a growth spurt before the rest of the body, leading to temporary disproportion.
3. **Correct answer – C**
   This male sex hormone triggers the development of reproductive organs and the initiation of sperm production.
4. **Correct answer – B**
   Hormonal changes cause these glands to produce excess oil, which clogs pores and leads to breakouts.
5. **Correct answer – C**
   This is the visible protrusion in the throat caused by the enlargement of the larynx in boys during puberty.
6. **Correct answer – C**
   Ovulation is the specific biological term for the monthly release of a mature egg from the ovary.
7. **Correct answer – C**
   Without fertilization, the body sheds the prepared uterine lining, resulting in a period.
8. **Correct answer – B**
   Menarche marks the beginning of a girl's reproductive phase during puberty.
9. **Correct answer – B**
   Adolescence is a time of significant brain development where individuals begin to think more critically and self-decisively.
10. **Correct answer – C**
    This is the typical age range when the menstrual cycle stops and a female is no longer reproductively active.

Worksheet – 2

1. **Correct answer – C**
   They are essential for bodybuilding and the repair and maintenance of body tissues.
2. **Correct answer – C**
   This period involves a high-speed expansion of the skeleton that requires more calcium than other life stages.
3. **Correct answer – B**
   Zinc is vital specifically for overall body growth and the process of sexual maturation.
4. **Correct answer – C**
   This method is recommended to maintain community hygiene and protect the environment.
5. **Correct answer – C**
   This project makes affordable, biodegradable pads available specifically through Jan Aushadhi Kendras.
6. **Correct answer – B**
   The Shuchi Scheme is a specific localized initiative implemented by the Karnataka state government.
7. **Correct answer – C**
   Trusted adults can help adolescents navigate technology in a positive and constructive way.
8. **Correct answer – C**
   These activities help young people stay in good physical shape through exercise and fresh air.
9. **Correct answer – B**
   The transitional nature of adolescence can lead to insecurity, making teenagers more susceptible to addiction.
10. **Correct answer – C**
    The HIV virus is a deadly pathogen that can be transmitted through the reuse of infected needles.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**
   Adolescents often look awkward because limbs typically lengthen before the rest of the body regains normal proportions.
2. **Correct answer – B**
   While girls often start their growth spurt earlier, both genders generally complete their height development by late adolescence.
3. **Correct answer – C**
   Identifying emotional confusion is the first step toward resisting peer pressure and making smart choices.
4. **Correct answer – C**
   Acne refers to the overall skin state, while pimples are the specific, visible results of excess oil and infection.
5. **Correct answer – B**
   Rising estrogen levels act as the biological signal that triggers the ovaries to release an egg.
6. **Correct answer – B**
   Guidance helps adolescents prevent unintended negative impacts that can come from the careless use of social media.
7. **Correct answer – A**
   Without a fertilized egg to support, the body naturally discards the prepared lining of the uterus.
8. **Correct answer – B**
   Beyond physical exercise, the environment of outdoor sports promotes overall fitness through fresh air.
9. **Correct answer – A**
   Without protein, the body lacks the specific building blocks needed to fix and sustain muscles and organs.
10. **Correct answer – B**
    This strategy uses the natural adolescent tendency to follow peers to promote positive health choices.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, C, D**
    A, **C**, and **D** are biologically accurate, statement **B** is incorrect because menarche is the *start* of menstruation, not the end.
12. **Correct answer – B, C**
    Estrogen and Progesterone coordinate the menstrual cycle and pregnancy, while estrogen specifically drives female secondary sexual characteristics like breast development.
13. **Correct answer – A, B, D**
    These specific programs are central government initiatives launched to improve access to hygiene products and health awareness across the country.
14. **Correct answer – A, C**
    These are secondary sexual characters specifically observed in girls during puberty as their bodies prepare for reproductive maturity.
15. **Correct answer – A, B, C**
    These outdoor activities and sports are recommended because they keep the body in good shape and promote overall health through fresh air and exercise.

ASSERTION AND REASONING TYPE

16. **Correct answer – D**
17. **Correct answer – A**
18. **Correct answer – C**
19. **Correct answer – A**

MATRIX MATCHING TYPE

20. **Correct answer – D**
    a-R, b-P, c-Q, d-S
21. **Correct answer – B**
    a-P, b-Q, c-R, d-S

COMPREHENSHION TYPE

Passage1

22. **Correct answer – C**
23. **Correct answer – B**
24. **Correct answer – C**

Passage2

25. **Correct answer – B**
26. **Correct answer – C**
27. **Correct answer – C**

4 Reproduction in Plants

Solutions

Worksheet – 1

1. **Correct Answer (C) Fruit**
   The fruit is the mature ovary of a flower that develops after fertilization to protect the seeds and help in their dispersal. Stems and roots are vegetative parts, and leaves are for photosynthesis.
2. **Correct Answer (B) Complete flower**
   A complete flower is defined by having all four major whorls: sepals (calyx), petals (corolla), stamens (androecium), and pistil (gynoecium).
   Incomplete flowers miss one or more of these parts.
3. **Correct answer (C) Pistil**
   The pistil (or carpel) is the female reproductive part; it contains the ovary, which houses ovules that eventually become seeds.
   The stamen is male. Corolla (petals) and calyx (sepals) are non-reproductive parts.
4. **Correct answer (C) Reproduction**
   Reproduction is the biological process by which plants produce new offspring or 22 "new life". Photosynthesis is food production, respiration is energy release, and transpiration is water loss.
5. **Correct Answer (A) Root**
   The roots are responsible for anchoring the plant firmly in the soil and absorbing water. The stem supports the plant above ground.
6. **Correct answer (A) Papaya and maize**
   Papaya and maize are examples of unisexual flowers, meaning an individual flower has either male or female reproductive organs, but not both. Rose, hibiscus, and lily are typically bisexual (hermaphrodite).
7. **Correct Answer (A) Calyx**
   The calyx (made up of sepals) protects the delicate inner parts of the flower while it is still a bud. Corolla attracts pollinators; androecium and gynoecium are the reproductive organs.
8. **Correct Answer (B) Whorl arrangement**
   These terms describe the position of the gynoecium (whorl) relative to the other floral parts on the thalamus.
9. **Correct Answer (C) Androecium & D) Gynoecium (Combination)**
   While you question asks for them "together," the male parts are the Androecium and the female parts are the Gynoecium. In a single choice format, this question usually refers to the "essential whorls."
10. **Correct answer (B) Calyx protects the bud, corolla attracts insects**
    The calyx provides protection during the bud stage, while the corolla (petals) uses bright colours and scents to attract pollinators like insects.

Worksheet-2

1. **Correct Answer (B) Offspring are clones of parent**
   Asexual reproduction involves a single parent, resulting in offspring that are genetically identical to the parent.
   Options A and C describe sexual reproduction; option D is incorrect as asexual reproduction is typically faster.
2. **Correct answer (B) Sweet potato**
   Sweet potatoes have tuberous roots that can develop adventitious buds to grow into new plants.
   Potato (stem tuber), Onion (bulb), and Bryophyllum (leaf) use different parts for propagation.

3. **Correct Answer (C) Sugarcane**
Sugarcane is commercially grown by planting stem cuttings that contain at least one "eye" or node. Bryophyllum propagates via leaves; Onion via bulbs; Sweet potato via roots.
4. **Correct Answer (D) Desirable qualities of two related plants**
Grafting is used to combine the hardy root system of one plant with the superior fruit or flower qualities of another. It does not involve unrelated plants or just soil and water.
5. **Correct Answer (A) Roots develop where stem touches soil**
In layering, a branch is bent and covered with soil while still attached to the parent, causing it to grow roots. B describes cutting, and C describes budding.
6. **Correct Answer (A) Scion**
The scion is the upper part of the graft that becomes the new shoot system. The stock provides the root system.
7. **Correct Answer (B) Budding**
Budding is a specific type of grafting where only a single bud is joined to the bark of the rootstock.
8. **Correct Answer (A) Stem tuber**
A stem tuber, like a potato, is an underground stem that swells with stored starch. Root tubers are modified roots; bulbs are modified leaves/stems.
9. **Correct Answer (A) Stolons**
Runners are horizontally growing stems that run along the soil surface and are scientifically called stolons.
10. **Correct Answer (B) Grass**
Most grasses and strawberries propagate through runners (stolon's) that take root at nodes to form new plants.

Worksheet-3

1. **Correct Answer (B) Anther to stigma**
Pollination is specifically the process where pollen grains are moved from the male anther to the receptive female stigma.
Options A, C, and D describe incorrect pathways or parts not involved in the initial pollen transfer.
2. **Correct Answer (C) Entomophily**
The technical term for pollination carried out by insects (like bees or butterflies) is entomophily.
Anemophily is by wind, Hydrophily is by water, and Zoophily is a broad term for any animal.
3. **Correct Answer (A) Water, oxygen, and suitable temperature**
These three factors are the essential environmental conditions required for a seed to break dormancy and begin to grow.
Sunlight is usually only needed after the plant has sprouted (germinated) to begin photosynthesis; soil and pollen are not absolute requirements for the germination phase itself.
4. **Correct Answer (B) Rose**
Roses have brightly colored petals and scents specifically to attract insects for pollination.
Rice, grass, and maize are typically pollinated by the wind (anemophilous).
5. **Correct answer (C) Water**
Hydrophily refers to pollination where pollen is distributed by water currents, common in some aquatic plants.
6. **Correct Answer (B) Wind**
Anemophily is the term for pollination by wind, which is common in cereal crops and grasses.
7. **Correct Answer (B) Fertilization**
After fertilization occurs, the flower's non-reproductive parts like sepals, petals, and stamens usually wither and fall off as the ovary develops into a fruit. Pollination is just the first step; these parts remain until the actual fusion (fertilization) is successful.
8. **Correct answer (C) Insects**
Flowers use bright colours and fragrances as signals to attract insects like bees, which seek nectar and accidentally transfer pollen in the

process. Wind and water do not respond to visual or scent cues.

9. **Correct Answer (B) Seed coat**
The seed coat (or testa) is the hard protective outer layer that shields the embryo from damage and dehydration. Cotyledons and endosperm provide food; the embryo is the "baby plant" inside.
10. **Correct answer (D) Pollen**
Pollen is required to create the seed through fertilization, but once the seed already exists, it does not need pollen to germinate (start growing). Water, oxygen, and temperature are the active requirements for the growth process.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (B) The shoot system contains leaves, which act as the "food factory" via photosynthesis.**
The shoot system includes leaves, which contain chlorophyll to capture sunlight and convert carbon dioxide and water into glucose (food). Roots generally lack chlorophyll and are located underground where there is no light for photosynthesis.
2. **Correct Answer (C) The Pistil**
Gynoecium is the formal botanical term for the female reproductive part of a flower, also commonly called the pistil or carpel. Stamen is the Androecium (male); Corolla and Calyx are non-reproductive parts.
3. **Correct Answer (B) Incomplete and Unisexual (Male)**
Because the flower lacks an ovary and stigma (female parts) but has stamens (male parts), it is unisexual. Since it is missing one of the four main whorls, it is also incomplete. Bisexual flowers would have both male and female parts.
4. **Correct Answer (A) Continuity of life on Earth**
The biological purpose of reproduction is to ensure that a species does not go extinct and continues to exist through successive generations.
5. **Correct Answer (B) Bryophyllum**
Bryophyllum is famous for vegetative propagation via "epiphyllous buds" that form along the margins of its leaves.
6. **Correct answer (B) Cutting allows for faster production and ensures the new plant is exactly like the parent.**
Cuttings are clones; they bypass the slow seedling stage and guarantee that the new plant will have the exact same flower color or fruit quality as the original.
7. **Correct answer (C) To combine desirable characteristics and improve varieties.**
Farmers use methods like grafting to combine the strength of one plant (like disease-resistant roots) with the high yield of another.
8. **Correct Answer (C) Radicle**
The radicle is the part of the embryo that develops into the primary root; it emerges first to ensure the seedling can absorb water immediately.
The plumule becomes the shoot (stem/leaves).
9. **Correct Answer (B) Wind pollination (Anemophily)**
Wind-pollinated flowers don't need to waste energy on bright colors, nectar, or scent because they don't need to attract animals. They are usually small and green.
10. **Correct Answer (C) Vallisneria**
Vallisneria (Tape grass) is a classic example of hydrophily, where it releases its male flowers to float on the water surface to reach the female flowers.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer A) Stigma, C) Style, and D) Ovary**
The Gynoecium is the female reproductive part of the flower. It consists of the stigma (the receptive tip), the style (the stalk), and the ovary (the base containing ovules).

12. **Correct answer B) Large quantities of pollen grains and C) Light and dry pollen**

    Anemophilous (wind-pollinated) flowers produce vast amounts of pollen because much of it is lost in the air. This pollen is light and dry so it can easily be carried by the breeze. Bright colors and nectar are used to attract insects, not wind.

13. **Correct Answer A) Papaya, C) Watermelon, and D) Maize**

    These plants produce unisexual flowers, meaning a single flower is either male (staminate) or female (pistillate), but not both. Rose is a bisexual flower containing both male and female parts.

14. **Correct Answer A) To ensure superior qualities, B) To combine root/fruit quality, and D) To produce improved varieties faster**

    Farmers use artificial propagation because it guarantees the new plant is a genetic clone of the high-quality parent, combines the best of two plants (like a strong rootstock), and yields fruit much faster than seeds.

    The "Monocot" claim is scientifically incorrect in this context.

15. **Correct Answer A) Roots with buds, C) Runners, and D) Leaf buds**

    These are natural methods where the plant reproduces on its own without human intervention (e.g., grass spreading via runners or Bryophyllum growing buds on leaves). Grafting is an artificial method performed by humans.

ASSERTION AND REASONING TYPE

16. **Correct Answer (A) Both Assertion (A) and Reason (R) are true, and (R) is the correct explanation of (A).**

    Because wind is an "abiotic" (non-living) agent, the direction it blows is random. To compensate for the high chance of pollen being lost, plants produce millions of grains to ensure at least a few land on a stigma.

17. **Correct Answer (A) Both Assertion (A) and Reason (R) are true, and (R) is the correct explanation of (A).**
    A flower is only "complete" if it has all four whorls. Since unisexual flowers (like Papaya) are missing either the male or female reproductive whorl, they are by definition incomplete.

18. **Correct Answer (D) (A) is false, but (R) is true.**
    Assertion (A) is false because cutting is an artificial method of propagation—it requires a human to perform the act of cutting and planting. However, Reason (R) is a true statement describing how the process works.

19. **Correct Answer (A) Both Assertion (A) and Reason (R) are true, and (R) is the correct explanation of (A).**
    These classifications are based specifically on the position of the ovary on the thalamus. If the ovary is above other parts, it's hypogynous; if it's below, it's epigynous.

MATRIX MATCHING TYPE

20. **Correct answer (A) a→P, b→Q, c→R, d→S**
21. **Correct answer (A) a→P, b→Q, c→R, d→S**

COMPREHENSION TYPE

Passage-1

22. **Correct Answer (B) Wind**
23. **Correct Answer (C) Androecium**
24. **Correct Answer (B) Fusion of gametes**

Passage-2

25. **Correct Answer (B) Eyes**
26. **Correct Answer (C) Cutting**
27. **Correct Answer (B) Roots, stems, or leaves**

PREVIOUS YEAR QUESTIONS

1. **Correct Answer (C) In normal conditions, all the flowers of Y will give rise to seeds after fertilization.**

   Plant X has unisexual flowers (either male or female), while plant Y has bisexual flowers. Because Y has both reproductive parts in the same flower, self-pollination and fertilization are highly likely, leading to seed formation.

2. **Correct Answer (B) Plant Y has unisexual female flowers only and female flowers cannot produce fruits in absence of pollinator.**

   Raghu covered the flowers, blocking pollinators. Plant X produced fruit, meaning it likely self-pollinated (bisexual). Plant Y produced no fruit because its female flowers needed pollen from a separate male flower, which was blocked by the cover.

3. **Correct answer (B) Petals are bright in color.**

   Wind-pollinated flowers do not need to attract insects, so they typically have small, green, or dull petals.

4. **Correct answer (C) X: Anther | Y: Mustard | Z: Carpel**

   X: Stigma and Style are female; Anther is male (Odd one).

   Y: Maize and Cucumber are unisexual; Mustard is bisexual (Odd one).

   Z: Sepal and Petal are non-reproductive; Carpel is reproductive (Odd one).

5. **Correct Answer (B) (iii), (iv) and (vi) only**

   (iii) Acacia, (iv) Bougainvillea, and (vi) Mimosa have features (scent, color, or nectar) that attract insects.

5 SOIL

Solutions

Worksheet – 1

1. **Correct answer – B**

   Humus is the dark, organic material formed by the decay of plant and animal matter; it is essential for soil fertility.

2. **Correct answer – C**

   Weathering is the slow, natural process that breaks down massive rocks into smaller particles to eventually form soil.

3. **Correct answer – D**

   A soil profile is like a "side view" of the earth, showing the different horizontal layers (horizons) from the surface down to the bedrock.

4. **Correct answer – B**

   The R Horizon (or Bedrock) is the solid, unweathered rock layer at the bottom of the soil profile that is extremely hard to dig.

5. **Correct answer – A**
   This top layer is rich in humus and minerals, making it soft and porous enough to hold the water and air needed for seeds to sprout and roots to grow.

6. **Correct answer – C**
   Sandy soil is made of large particles that don't fit closely together, resulting in large spaces that allow for high aeration but poor water retention.

7. **Correct answer – B**
   Because clay particles are so small and pack tightly, they trap and hold a lot of water in their tiny spaces, making the soil heavy.

8. **Correct answer – C**

   Loam is a balanced mixture of sand, clay, and silt. It has the right water-holding capacity and enough air for healthy plant growth.

9. **Correct answer – D**

   Percolation refers specifically to the downward movement of water through the pores of the soil.

10. **Correct answer – B**

    Using the formula

    Rate = Amount of water/ Time,

    we get, 100 mL /25 min = 4 mL/min

Worksheet – 2

1. **Correct answer – C**

   Sandy soil has large particles and high porosity, which allows water to drain quickly and prevents waterlogging.

2. **Correct answer – B**

   Fine clayey soils hold the moisture and organic nutrients that cereal crops like wheat need for growth.

3. **Correct answer – A**

   The statement specifically lists wind and rainfall as essential climatic factors that affect the soil profile and bring changes to its structure.

4. **Correct answer – B**

   The statement states that while natural factors cause erosion, human activities accelerate the damage to the land much more severely.

5. **Correct answer – D**

   Roots act as a biological net that anchors the topsoil, making it resistant to being washed or blown away.

6. **Correct answer – C**

   Growing pulses after paddy replaces nitrogen and other minerals that were depleted by the previous crop.

7. **Correct answer – A**

Loamy soil is a balanced mixture of sand, silt, and clay, providing the perfect combination of drainage and nutrients for pulses.

8. **Correct answer – C**

Soil pollution specifically refers to the contamination of the earth by toxic chemicals.

9. **Correct answer – A**

Paddy requires standing water to grow, so soil that holds water (low percolation) rather than letting it drain away is ideal.

10. **Correct answer – B**

Erosion is the physical process of wearing away the Earth's surface by natural agents like wind, water, or ice.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**

Clay particles are very fine and pack tightly together, allowing them to hold more water than sandy or loamy soils.

2. **Correct answer – A**

Percentage of moisture = Loss in weight (15g)/Original weight (150g) ×100

% water = [(150 - 135)/150] × 100 = 15/150 × 100 = 10%

3. **Correct answer – D**

On hot days, moisture in the soil evaporates; this water vapor makes the air above the land denser, causing sunlight to reflect and shimmer.

4. **Correct answer – C**

The volume of water absorbed is the difference between what you started with (U) and what leaked through into the cylinder (V).

5. **Correct answer – A**

Clay has very small air spaces and high-water retention, which leads to waterlogging and "heavy" soil conditions.

6. **Correct answer – B**

Volume absorbed = 100 - 70 = 30 ml

Weight of soil = 50 g, Volume water absorbed = 30 mL → % water = (30/50) × 100 = 60%

7. **Correct answer – C**

Vegetation acts as a "skin" for the earth; without it, rain shatters the soil structure and washes it away easily.

8. **Correct answer – A**

Deforestation is the removal of trees, which leaves soil exposed and loose, directly causing erosion rather than preventing it.

9. **Correct answer – B**

While silt is a component of soil, the three primary classifications discussed in this context are Sandy, Clayey, and Loamy.

10. **Correct answer – C**

Monoculture exhausts specific nutrients; pulses restore nitrogen and break the cycle of degradation.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, B, C**

Air, water, and soil are naturally occurring substances essential for life, whereas plastic is a synthetic human-made material.

12. **Correct answer – A, B, C**

Absorption varies by texture; clayey soil absorbs and retains the most water, while sandy soil allows most of it to percolate through.

13. **Correct answer – B, D**

Large-scale production of paper and plywood requires cutting down trees, leading to deforestation and subsequent soil erosion.

14. **Correct answer – A, C**

Jute and cardboard are biodegradable organic materials, whereas acid rain and plastics introduce harmful chemicals/pollutants into the soil.

15. **Correct answer – A, C, D**

Erosion turns fertile land into desert (desertification), causes water runoff that leads to floods, and ruins crops, which can cause famine.

ASSERTION AND REASONING TYPE

16. **Correct answer – D**

Clayey soil is actually "heavy" because it holds a lot of water; however, the reason accurately describes why it has little air.

17. **Correct answer – A**

Evaporating water vapor creates a physical lens effect that causes the "shimmering" visual on hot days.

18. **Correct answer – B**

While soil is a filter, its persistence (staying toxic) is due to its static nature and lack of "flow" compared to air or water.

19. **Correct answer – C**

Human activities like deforestation actually have a much larger and more constant impact on soil erosion than rare natural calamities.

MATRIX MATCH TYPE

20. **Correct answer – A**

a-R, b-T, c-P, d-Q, e-S, f-U

21. **Correct answer – D**

a-R, b-P, c-Q, d-S

COMPREHENSHION TYPE

Passage1

22. **Correct answer – B**
23. **Correct answer – B**
24. **Correct answer – C**

Passage2

25. **Correct answer – B**
26. **Correct answer – B**
27. **Correct answer – B**

Previous Year Questions

1. **Correct answer – C**

Humus – Clay – Sand – Gravel Soil settles in layers with humus on top, then clay, sand, and gravel at the bottom.

2. **Correct answer – A**

W Rich in humus- The top layer (W) of soil contains the most humus.

3. **Correct answer –B**

Soil Q absorbs the most water (highest weight gain), making it best for shaping statues.

6 Our forest

Solutions

Worksheet – 1

1. **Correct Answer (B) One-third**

   Forests cover approximately 31% of the Earth's land surface, which is roughly one-third.

   Options A, C, and D are numerically significantly higher or lower than the current global average.

2. **Correct Answer (C) Emergent layer**

   The emergent layer is the topmost layer, consisting of the very tallest trees that poke above the rest of the forest. The canopy is the dense ceiling below it, the understorey is in the middle, and the forest floor is the ground.

3. **Correct Answer(D) Canopy**

   The canopy acts as a dense "roof" or umbrella over the forest floor.

   The crown layer refers to individual tree tops; the emergent zone is above the "roof," and the understorey is below it.

4. **Correct answer (C) Understorey**

   The understorey (or shrub layer) is the vegetation layer located between the canopy and the forest floor.

   The crown and canopy are above this layer; the emergent layer is the highest point.

5. **Correct answer (C) Production of plastic materials**

   Forests do not produce plastic; plastics are synthetic materials typically made from polymers derived from petroleum. Forests actively supply medicinal plants, regulate gases like oxygen/CO2, and provide industrial employment.

6. **Correct Answer(B) Between trees and herbs**

   In the vertical structure of a forest, shrubs typically occupy the space between the tall trees above and the small herbs on the ground.
   Shrubs are too short to be above trees or in the emergent layer, and they are distinct from the herbs growing directly on the forest floor.

7. **Correct Answer(B) Beekeeping**

   Apiculture is the scientific term for the maintenance of bee colonies, commonly for honey production.
   Fish farming is aquaculture; silk weaving is sericulture.

8. **Correct answer (C) Food, fodder, and firewood**

   Local communities rely on forests for a wide range of essential resources including food, animal fodder, and fuel for cooking.
   While forests provide oxygen, rainfall, and medicine, they are not the only benefits provided.

9. **Correct answer(B) Soil fertility**

   Decaying organic matter (humus) from(leaves adds vital nutrients back into the ground, significantly increasing soil fertility. Decaying leaves have no significant impact on air temperature, wind speed, or the intensity of sunlight.

10. **Correct option (A) Differences in species**

    Tree crowns vary in shape primarily due to the specific biological characteristics of different species, though climate also plays a role.

    Climate, soil, and sunlight vary greatly across the globe, and these variations actually cause different species to thrive with different crown shapes.

Worksheet -2

1. **Correct Answer(B) Western coast of India**
   Tropical evergreen forests thrive in areas with very high rainfall (over 200 cm), making the Western Ghats (Western coast) a primary location.
   The Thar Desert is arid; the Deccan Plateau and Indo-Gangetic Plain primarily feature deciduous or thorn forests.
2. **Correct Answer (A) Deodar**
   Deodar is a classic coniferous tree found in the temperate altitudes of the Himalayas.
   Coconuts are tropical/coastal; date palms are desert-dwelling; neem is found in tropical deciduous/dry regions.
3. **Correct Answer (C) Planting trees where forests were cut down**
   Reforestation specifically refers to restoring forest cover in areas where it previously existed but was removed.
   Planting on barren land where there was no previous forest is afforestation.
4. **Correct Answer (C) Ecological imbalance**
   Deforestation disrupts food chains, water cycles, and carbon levels, leading to widespread ecological imbalance.
   Deforestation actually reduces rainfall and soil fertility while increasing global temperatures.
5. **Correct Answer (D) 80%**
   Moist tropical forests (rainforests) maintain very high humidity, often averaging around 77% to 88% year-round.
   Options A, B, and C are too low for a "moist tropical" classification.
6. **Correct answer (B) Nilgiris**

   In Southern India, temperate forests (locally known as Sholas) are found in the higher reaches of the Nilgiris, Anaimalai, and Palani hills.

   The Aravalli, Vindhya, and Satpura ranges are generally lower in altitude and characterized by deciduous or scrub vegetation.
7. **Correct Answer (B) Chipko Andolan**

   The Chipko Movement (from the word 'hug') involved villagers literally hugging trees to prevent loggers from cutting them down.

   The Narmada Bachao Andolan is about dams; Silent Valley is about a specific hydroelectric project in Kerala.
8. **Correct Answer(B) March 1973**

   The modern movement famously began in March 1973 in the Chamoli district of Uttarakhand.

   The other dates do not align with the start of this specific environmental campaign.
9. **Correct Answer (D) Chamoli**

   Gopeswar is the administrative headquarters of the Chamoli district, the birthplace of the Chipko Movement.

   Almora, Tehri, and Pauri are neighboring districts but not the location of Gopeswar.
10. **Correct Answer (B) Floods**

    Without tree roots to anchor soil and absorb water, heavy rain leads to rapid runoff, causing frequent floods.
    Deforestation decreases biodiversity, increases soil erosion, and leads to a warmer climate (global warming).

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (B) WWF and UNESCO**

   These organizations provide global frameworks and guidelines for the protection of biodiversity and world heritage sites.

   WHO/UNICEF focus on health/children; WTO/World Bank focus on trade/finance; FAO/Greenpeace involve food and activism but UNESCO/WWF are standard academic answers for protection guidelines.

2. **Correct Answer(C) It showed that local communities could take the lead in conservation without waiting for official government action.**
   It is celebrated as a "people's movement" because it was grassroots, showing that ordinary citizens could protect their own environment.
   It was not about high-tech machinery, industrial development, or primary funding from international groups.
3. **Correct Answer (C) The Canopy**
   The canopy is the thick layer where branches and leaves of different trees overlap to form a continuous "roof."
   The emergent layer is above the roof; the understorey and floor are below it.
4. **Correct Answer (C) Chipko Andolan**
   Sunderlal Bahuguna was a prominent leader of this 1973 movement, famous for his slogan "Ecology is permanent economy."
   Save Silent Valley was in Kerala; Narmada Bachao is about dams; Appiko was a later movement in Karnataka inspired by Chipko.
5. **Correct answer (B) Parks protect existing ecosystems, while nurseries provide resources for new planting.**
   National Parks are for conservation of wildlife and habitats, whereas nurseries are used to grow saplings for reforestation.
   Neither is used primarily for industry or tourism exclusively.
6. **Correct Answer(B) Bamboo, Aini, and Rosewood**
   These are typical species found in the high-rainfall Moist Tropical Evergreen forests of India.
   Sal and Acacia are deciduous/dry forest species; Oak and Deodar are temperate mountain species.
7. **Correct option (C) Alpine Forest**
   Found at very high altitudes in the Himalayas, Alpine forests experience extreme cold similar to arctic conditions.
   Tropical and subtropical forests are found in much warmer climates.
8. **Correct option (B) Maintain balance of atmospheric gases**
   Through photosynthesis, forests absorb carbon dioxide and release oxygen, maintaining the vital gas balance.
   Forests actively work against plastic production, mining, and desertification.
9. **Correct option (B) Recycling nutrients**
   Decomposers (like fungi and bacteria) break down dead matter, returning nutrients to the soil for plants to reuse.
   They increase soil fertility and aid seed germination by clearing space and enriching soil.
10. **Correct option (C) Hugging trees to prevent cutting**
    The defining feature of the movement was people physically placing themselves between the trees and the saws.
    While legal action and planting occur in conservation, "hugging" was the primary method that gave the movement its name.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct option A, C, and D**
    A: Forest nurseries provide the saplings needed for reforestation efforts.
    B: Overgrazing is harmful as it destroys young saplings and leads to
    D: Reducing pollution protects the overall health of the forest ecosystem.
12. **Correct option A, B, C, and D**
    A: Sunderlal Bahuguna was a key leader who popularized the movement.
    B: It officially launched in March 1973 in Chamoli (Uttarakhand).
    C: The name "Chipko" comes from the act of hugging trees.
    D: Women, like Gaura Devi, were instrumental in the movement's success.
13. **Correct option A, B, and D**
    A: The emergent layer is the highest and gets direct, maximum sunlight.
    B: The canopy is defined by the dense, overlapping crowns of trees.D: The forest

floor is where leaf litter and organic matter decay.

14. **Correct option A, B, and D**
A: These forests are characterized by high rainfall, typically 200 cm or more.
B: High humidity (often 77–88%) is a defining trait.
D: The high moisture allows trees to remain green year-round.
15. **Correct option A, B, and C**
A: Decomposers (fungi, bacteria) break down dead plants and animals.
B: This process returns essential nutrients to the cycle.
C: By adding organic matter (humus), they significantly enrich the soil

ASSERTION AND REASONING TYPE

16. **Correct Answer (A) Both Assertion and Reason are true, and Reason is the correct explanation. (The movement is "grassroots" specifically because it was driven by the local community.)**
17. **Correct answer (A) Both Assertion and Reason are true, and Reason is the correct explanation. (Planting trees on new land directly contributes to the conservation and expansion of forest cover.)**
18. **Correct answer (A) Both Assertion and Reason are true, and Reason is the correct explanation. (In dry tropical climates, trees shed leaves to reduce transpiration and save water during the dry season.)**
19. **Correct answer (A) Both Assertion and Reason are true, and Reason is the correct explanation. (The physical damage to saplings and the exposure of soil are the direct ways grazing harms the ecosystem.)**

MATRIX MATCH TYPE

20. **Correct Answer (A) a→P, b→Q, c→R, d→S**
21. **Correct Answer (A) a→P, b→Q, c→R, d→S**

COMPREHENSION TYPE

Passage-1

22. **Correct Answer (B) Regulating temperature**
23. **Correct Answer (C) Stratosphere**
24. **Correct Answer (C) Canopy**

Passage -2

25. **Correct answer (C) Very high-altitude**
26. **Correct answer (C) Maintaining ecological balance**
27. **Correct answer (C) Hilly regions**

Previous Year Questions

1. **Correct answer (D) All of these**

Forests are a vital source of diverse materials including timber, medicinal plants, and resins and gums.

2. **Correct option (B) (i), (ii) and (iv) only**

(i) True: Lightning is a common natural cause of forest fires.

(ii) True: The understorey consists of vegetation growing in the shade beneath the canopy.

(iii) False: In the absence of forest cover, soil erosion increases because there are no roots to bind the soil.

(iv) True: Humans rely on forests for industrial raw materials like timber, resin, and lac.

(v) False: The canopy receives the most sunlight (along with the emergent layer), while the forest floor receives the least.

3. **Correct answer (D) All of these**

The absence of forests leads to multiple environmental crises:

A: CO_2 levels rise because there are fewer trees to perform photosynthesis.

B: Without roots to absorb water and trees to regulate the water cycle, floods and droughts become more frequent.

C: Loss of habitat directly causes the extinction of species that depend on the forest for survival.